OBJECT ORIENTED SOFTWARE ENGINEERING

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Requirements Analysis

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- **Requirements Analysis result in:**
  - Specifying software’s **operational characteristics**
  - indicates software's interface with other system elements
  - establishes **constraints** that software must meet

- Requirements analysis allows you to elaborate on basic requirements established during the inception, elicitation, and negotiation tasks that are part of requirements engineering

The requirements modeling action results in one or more of the following types of models:

- **Scenario-based models** of requirements from the point of view of various system “actors”
- **Data models** that depict the information domain for the problem
- **Class-oriented models** that represent object-oriented classes (attributes and operations) and the manner in which classes collaborate to achieve system requirements
- **Flow-oriented models** that represent the functional elements of the system and how they transform data as it moves through the system
- **Behavioral models** that depict how the software behaves as a consequence of external “events”

A Bridge

Requirements analysis model - These models provide a software designer with information that can be translated to architectural, interface, and component-level designs.

- Finally, the requirements model provides the developer and the customer with the means to assess quality once software is built.

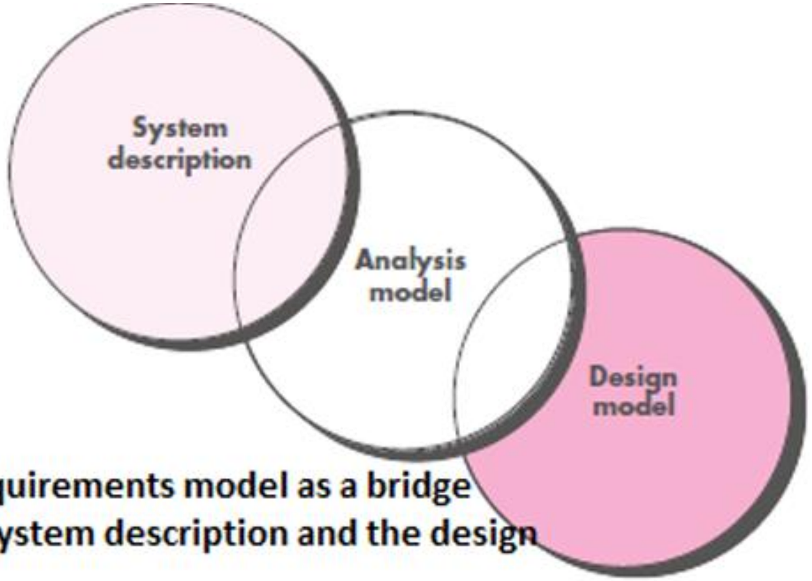
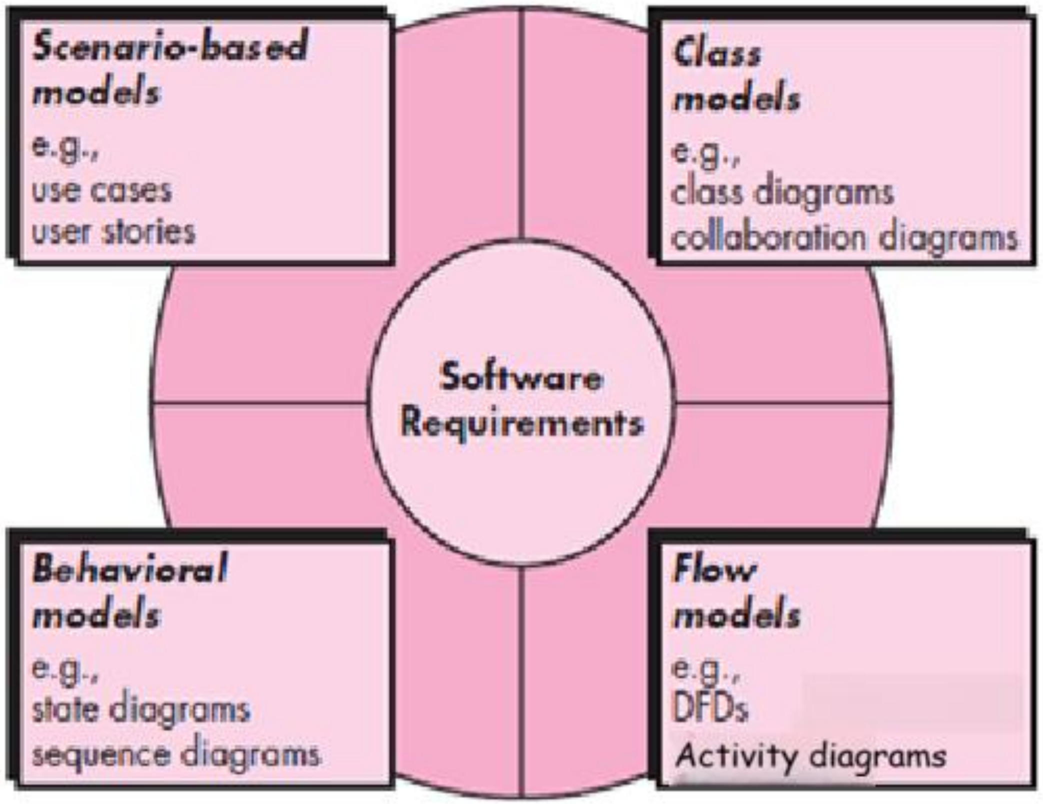


Fig 6.1: The requirements model as a bridge between the system description and the design model

Following **steps are performed** in generating Analysis models:

1. Define the **domain** to be investigated.
2. Collect a **representative sample of applications** in the domain.
3. **Analyze each application** in the sample.
4. Develop an **analysis model** for the objects.

Elements of Requirements Analysis



Elements of Analysis model

The intent of the **analysis model** is to provide a **description of the required informational, functional, and behavioral domains** for a computer-based system.

Elements of the Requirements (Analysis) Model

1. **Scenario-based elements**
2. **Class-based elements**
3. **Behavioral elements**
4. **Flow-oriented elements**

Elements of Analysis Model

1. Scenario-based elements

Describe the system from the **user's point of view** using **scenarios** that are depicted in **use cases** .

2. Class-based elements

Identify the **domain classes** for the objects manipulated by the actors, the **attributes** of these classes, and how they interact with one another; they utilize **class diagrams** to do this.

3. Behavioral elements

Use **state diagrams** to represent the state of the system, the **events** that cause the **system to change state**, and the actions that are taken as a result of a particular event; can also be applied to each class in the system and **Sequence diagrams** to represent the interaction among the system's objects.

4. Flow-oriented elements

Use **data flow diagrams** to show the **input data** that comes into a system, what **functions** are applied to that data to do transformations, and what resulting **output data** are produced and use Activity diagrams to represents the control flow within the system components.

THANK YOU

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